Functions, $\boxed{\S}$12

1. Which of the following functions are one-to-one (injective); which are onto (surjective)?

(a) $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$

(b) $g : \mathbb{R} \to \mathbb{R}_{\geq 0}$ given by $g(x) = x^2$

(c) $h : \mathbb{R} \smallsetminus \{0\} \to \mathbb{R}$ given by $h(x) = 1 + \frac{1}{x}$

(d) $r : \mathbb{R} \to \mathbb{R}$ given by $r(x) = 3x^3 - 2$

2. Consider the function $f : \mathbb{Z}^2 \to \mathbb{Z}^2$ given by $f(m, n) = (m + n, m + 2n)$.

(a) Show that f is an injective (that is, one-to-one) function.

(b) Show that f is a surjective (that is, onto) function.

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3. A function $f : S \to T$ is called a *constant function* if there exists a $t \in T$ such that for all $s \in S$ we have $f(s) = t$.

(a) Is the following definition equivalent? Why or why not?

> A function is called a *constant function* if for all $s \in S$ there exists a $t \in T$ such that $f(s) = t$.

Hint: It isn't.

For the rest of this problem, let $g : S \to T$ and $h : T \to S$ be functions, and assume their composition $h \circ g$ is a constant function.

(b) Write out a well-quantified statement to explain what "assume their composition $h \circ g$ is a constant function" means.

(c) What can you say about h given that g is onto? Express your claim as an if-then statement, and prove it.

(d) What can you say about g given that h is one-to-one? Express your claim as an if-then statement, and prove it.

(e) Is it necessarily true that $g \circ h$ is a constant function?